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Contents

Intellectual Property Rights 4
Foreword 4
Modal verbs terminology 4
Introduction 4
1 Scope 5
2 References 5
2.1 Normative references 5
2.2 Informative references 5
3 Definition of terms, symbols and abbreviations 6
3.1 Terms 6
3.2 Symbols 6
3.3 Abbreviations 6
4 PAS contextual introduction 6
4.1 Background 6
4.2 Service Approach 6
4.3 Parking place organizational aspects 8
5 PAS functional specification 9
5.1 PAS in the ITS station 9
5.2 PAS functionality 10
5.3 Interfaces of the PAS 11
6 POIM-PA dissemination 13
6.1 POIM-PA dissemination concepts 13
6.1.1 POIM-PA dissemination requirements 13
6.1.2 Inclusion of POI information blocks 13
6.1.3 POIM-PA assembly 13
6.1.4 Trigger, update, repetition and termination of dissemination 13
6.2 POIM-PA dissemination constraints 14
6.2.1 Security constraints 14
6.2.1.1 Introduction 14
6.2.1.2 Service Specific Permission (SSP) 14
6.2.2 General priority constraints 14
7 POIM-PA Specification 15
7.1 POIM-PA structure 15
7.1.1 General Structure of a POIM-PA PDU 15
7.1.2 ITS PDU header 16
7.1.3 Management container 16
7.1.4 General Parking Place container 16
7.1.5 Parking Status container 16
7.1.6 Parking Supply Capabilities container 17
7.1.7 Related POI container 17
7.2 POIM-PA format 17
Annex A (normative): ASN.1 modules 18
History 19

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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Intelligent Transport Systems (ITS).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Introduction

The present document specifies the Parking Availability Service (PAS) and how an infrastructure ITS Station (ITS-S) can disseminate information about the static and dynamic status of a parking place and its offered parking spaces. The PAS increases awareness of parking options by sharing information about the availability of parking spaces corresponding to the relevant vehicle type and locally related services.

The PAS operates based on authoritative information from the applicable parking provides, which may be a private company, a municipality or a state-owned organization. Therefore, it is intended that the information has a high level of trustworthiness, compared to other solutions based on crowdsourcing.

The PAS is part of the generic class of POI services that uses the generic POIM Format to disseminate information about a specific POI, which in this case is a parking place.

1 Scope

The present document provides the specification of the Release 2 Parking Availability Service which disseminates dynamic parking availability information in various transport environments (on street, off street, in parking facilities and park & ride stations).

More specifically, the present document specifies the syntax and semantics of the Point Of Interest Message for Parking Availability (POIM-PA) based on the generic Point Of Interest Message (POIM) Format.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 102 894-2: "Intelligent Transport Systems (ITS); Facilities Layer; Part 2: Common Data Dictionary (CDD); Release 2".

[2] ETSI TS 103 097: "Intelligent Transport Systems (ITS); Security; Security header and certificate formats; Release 2".

[3] Recommendation ITU-T X.691/ISO/IEC 8825-2: "Information technology - ASN.1 encoding rules: Specification of Packed Encoding Rules (PER)".

[4] ETSI TS 102 940: "Intelligent Transport Systems (ITS); Security; ITS communications security architecture and security management; Release 2".

[5] ETSI TS 104 258: "Intelligent Transport Systems (ITS); Facilities Layer; Point of Interest Message (POIM) Format; Release 2".

[6] ETSI TS 102 965: "Intelligent Transport Systems (ITS); Application Object Identifier (ITS-AID); Registration; Release 2".

2.2 Informative references

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI TR 103 902: "Intelligent Transport Systems (ITS); ITS Framework; Terms, Symbols and Abbreviations; Release 2".

[i.2] TS 16157-6: "Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 6: Parking publications" (produced by CEN).

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ETSI TR 103 902 [i.1] and TS 16157-6 [i.2] apply.

3.2 Symbols

For the purposes of the present document, the following symbols apply:

PA_Trigger	Interface provided by the PAS for controlling POIM dissemination
PA_DataProvision	Interface provided by the PAS for making collected POIMs available
PA_Control	Interface required by the PAS to management plane entity(ies)
PA_DataOut	Interface required by the PAS to disseminate POIMs
PA_DataIn	Interface required by the PAS to gather POIMs and other service data
PA_Security	Interface required by the PAS to security plane entity(ies)

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TR 103 902 [i.1] apply.

4 PAS contextual introduction

4.1 Background

Efficient parking management is a key enabler for sustainable, safe and efficient road transport systems. Parking-related traffic constitutes a significant share of overall traffic demand, particularly in urban areas, where vehicles searching for parking spaces contributes to congestion, emissions, and noise pollution. Parking information is therefore of high relevance for traffic management authorities, infrastructure operators, service providers, and road users alike. In addition to parking attributes, information of point of interests (i.e. toilets, hotels other) can be linked to the parking facilities. Moreover, by adding a cooperative approach between infrastructure and vehicles, new services can be created like queue parking or parking sport reservation.

4.2 Service Approach

Parking management follows a structured service approach that distinguishes between two operational paths: an informative part, which is enabled by PAS and is in scope of the present document, and a cooperative part which is not in scope of the present document. In the **informative path**, where the infrastructure acts as an information provider and disseminates parking-related data to road users and service consumers without any direct feedback or information channel from the users back to the parking management system. In this mode, PAS supports the delivery of descriptive and status information about parking facilities without direct interaction or coordination with individual vehicles. The informative path forms the foundation for all parking-related information services and may be used independently or as an input for cooperative extensions, see Figure 1.

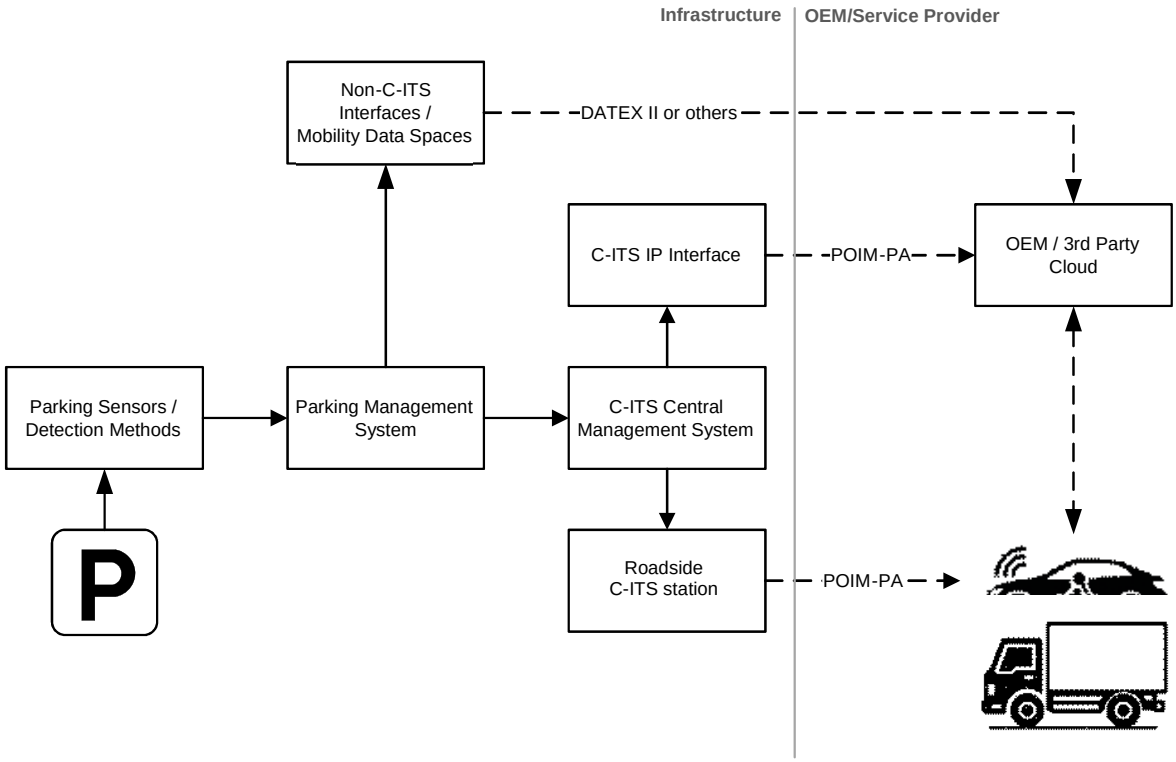


Figure 1: PAS informative path

- **Parking sensors and detection** methods capture dynamic parking data, primarily related to occupancy and availability. It could also be used to detect wrong or dangerously parked vehicles and check, if certain areas and spots are accessible. Detection may be performed using technologies such as entry and exit counting, camera-based systems, AI-supported video analytics, radar or lidar sensing, or manual validation based on visual observation. The selected detection approach significantly influences the accuracy, timeliness, and overall quality of the provided information.

- The **Parking Management System** manages one or more parking facilities and acts as the central aggregation point for parking-related data. It maintains static information, processes dynamic data received from the detection systems and derives additional indicators such as utilization statistics or trends. The system ensures data consistency and provides validated parking information to various dissemination channels.

- **Non-C-ITS interfaces / Mobility Data Spaces** enable backend-based data exchange with external systems and platforms, including Mobility Data Spaces and traffic or mobility service providers. Using standardized mechanisms such as DATEX II or comparable interfaces, parking information is shared for further processing, integration, and wide-area distribution.

- The **C-ITS Central Management System** receives parking information from the Parking Management System and prepares it for dissemination. It manages lifecycle, configuration, and consistency aspects required for the generation and distribution of C-ITS messages related to parking availability.

- The **C-ITS IP interface** supports IP-based communication between infrastructure systems and OEMs, third-party service provides or other C-ITS IP-based systems, for instance to share information from cities to highways. It enables the centralized distribution of parking information to backend systems, which may further aggregate and disseminate the data to connected vehicles.

- The **Roadside C-ITS station** provides short-range wireless communication between infrastructure and vehicles. Parking availability information is broadcast locally to vehicles within communication range, supporting low-latency and location-specific information delivery.

- On the **OEM** or **3rd Party service cloud**, parking information is processed within central cloud platforms or received directly by vehicles. The data is adapted to vehicle, service, or user-specific requirements and distributed accordingly, ensuring that end users receive relevant and up-to-date parking availability information.

The **cooperative path** extends the informative capabilities by enabling interactive management processes between vehicles or service providers and infrastructure. This cooperative approach allows the optimization of parking-related procedures, such as parking reservation services, which are built upon the information provided by the informative path. In such scenarios, vehicles may issue requests to the infrastructure and receive dedicated feedback messages related to these requests, enabling controlled and coordinated parking management, see Figure 2. This interaction pattern is conceptually comparable to established cooperative ITS mechanisms, such as the separation of information dissemination and further transactional exchanges ("agreements") as exemplified by the informative SPATEM/MAPEM vs. the cooperative SREM/SSEM message pairs in signalized intersections.

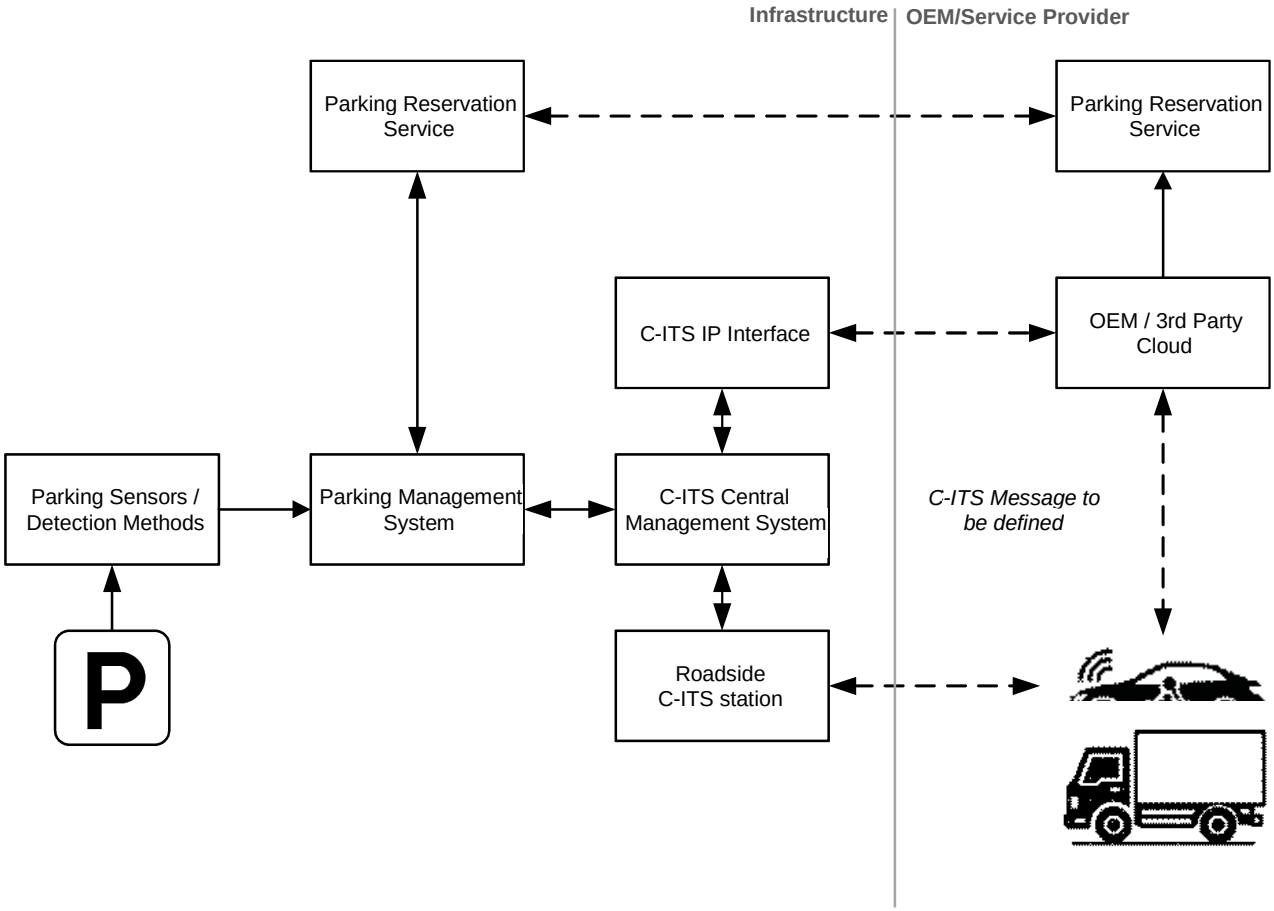


Figure 2: PAS cooperative path

4.3 Parking place organizational aspects

Parking places can differ enormously in location, size, orientation, purpose and more. Hence, it is important to introduce a nomenclature flexible enough to be used on many occasions but still clear enough to be used in a standardized way. For PAS, three layers - consisting of the parking facility, area and space - are used to describe parking places, see Figure 3:

- **Parking Facility** represents the highest level in the hierarchy and reflects the overall parking entity. A facility may correspond to a single physical structure (e.g. a parking garage), a service area along a motorway, or a logically grouped set of parking zones within an urban environment (e.g. multiple adjacent streets or districts). At this level, aggregated information can be provided, such as overall occupancy rates, availability trends, general access conditions, and basic descriptive metadata (e.g. name, address, opening hours).

- **Parking Areas** constitute an intermediate level of granularity and represent subdivisions of a parking facility. A parking area groups together parking spaces that share common characteristics, such as permitted vehicle types (e.g. passenger cars, trucks), spatial arrangement (e.g. parallel, perpendicular), orientation, or usage restrictions. Areas enable a more refined description of supply and demand within a facility, allowing differentiated occupancy information and constraints to be communicated for subsets of the overall parking infrastructure.

- **Parking Spaces** represent the highest level of detail and correspond to individual parking spots. Depending on the required level of granularity and data availability, parking spaces may be described with basic or detailed information. Basic descriptions typically include identifiers, status (e.g. free, occupied), and optional relative positions. Detailed descriptions may further specify geometric boundaries, orientation, occupancy rules, access relationships, reservation conditions, or additional attributes such as associated charging infrastructure. This level allows precise, space-specific information to be conveyed, as reflected by the *ParkingSpaceBasic* and *ParkingSpaceDetailed* data structures.



Figure 3: Example parking place organization

This hierarchical approach enables scalable information exchange and flexible implementation using data available: from facility-level summaries to space-level details. It is reflected in the data model defined in the present document, where aggregated and detailed status information can coexist and be provided according to system capabilities and use-case requirements.

5 PAS functional specification

5.1 PAS in the ITS station

The PAS is a facilities layer service that is based on the abstract concept of a generic POI service and which manages the generation, dissemination and reception of parking availability messages from the infrastructure to Vehicle or Personal ITS-S. Figure 4 illustrates a high level functional architecture of the PAS within the ITS station. The POIM-PA (see clause 7) is a facilities layer PDU that is exchanged among ITS-Ss.

The payload is generated by ITS applications in the transmitting ITS-S specifically for parking availability. At the transmitting ITS-S, the dissemination of a message is triggered by applications or by forwarding mechanisms. For this purpose, the applications may connect to other entities of the facilities layer such as other POI services or Infrastructure service, in order to collect relevant information for the generation of the payload. Once the message is generated, the service may repeat the dissemination, until the applications request the termination of the dissemination, or trigger another request to generate updated messages.

At the receiving ITS-S, the messages are processed by the services, and the content of the message is delivered to applications or to other facilities layer entity. In one typical application, the message is transmitted by a Roadside ITS-S and disseminated to a Vehicle ITS-S within a target destination area, in which the information included in the message is considered as relevant to traffic participants.

PAS may interact with other facilities layer entities, in particular:

- The Local Dynamic Map (LDM) service which is a facilities layer database at the receiving ITS-S may be updated with information from collected POIMs. ITS-S applications may then retrieve this information from the LDM for further processing.
- The Position and Time (PoTi) service which provides estimates of the kinematic state of the ITS-S and on time information to the sending PAS.

- The facilities layer Resource Management services to manage the dissemination of POIM.
- The facilities layer Security services to manage the security aspects of the POIMs.

The PAS may interface directly with lower layer for exchanging of POIMs with other ITS-Ss.

The functionalities of the PAS are defined in clause 5.2, the interfaces in Figure 4 are defined in clause 5.3.

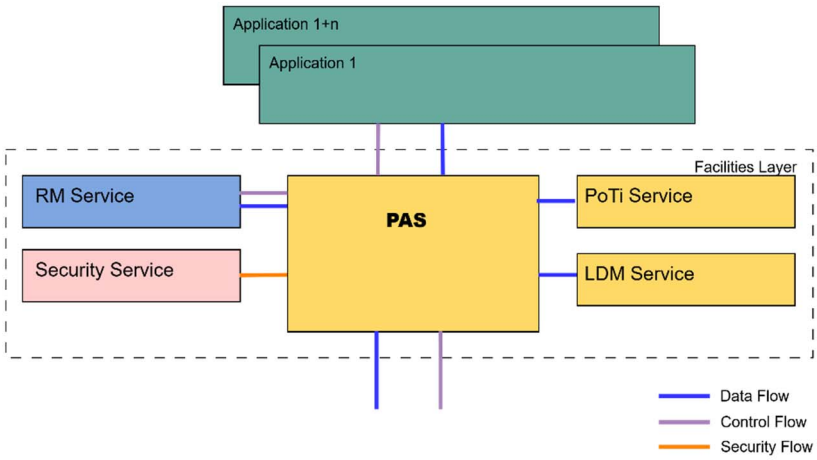


Figure 4: PAS and its logical interfaces

5.2 PAS functionality

The disseminating PAS and shall provide the following sub-functions:

- **Encode POIM:** This function constructs the POIM according to the format specified in Annex A.
- **POIM Dissemination Management:** This function implements the protocol operation of the originating ITS-S, as specified in clause 6, including in particular:
  - Activating the *Encode POIM* function to generate the POIM.
  - Determination of the POIM generation frequency.
  - Activation and termination of POIM dissemination by passing the POIM to either another facility layer entity such as Resource Management (RM) or a lower layer functionality such as the Networking & Transport Layer (NTL) for transmission.

The collecting PAS shall provide the following sub-functions:

- **Decode POIM:** This function decodes the received POIM.
- **POIM Collection Management:** This function implements the protocol operation of the receiving ITS-S, as specified in clause 6, including in particular:
  - The gathering of POIM-PA from either another facilities layer entity such as Resource Management or a lower layer functionality such as the Networking & Transport Layer (NTL).
  - Activating the *Decode POIM* function to decode the collected POIM.
  - Provision of the received POIM data to LDM and/or ITS applications of the receiving ITS-S.
  - Optionally, checking the information of received POIMs.

Figure 5 illustrates the component diagram with the functional blocks of the PAS and interfaces to other facilities and layers, which are detailed in the following. The interfaces to other entities and layers are defined in clause 5.3.

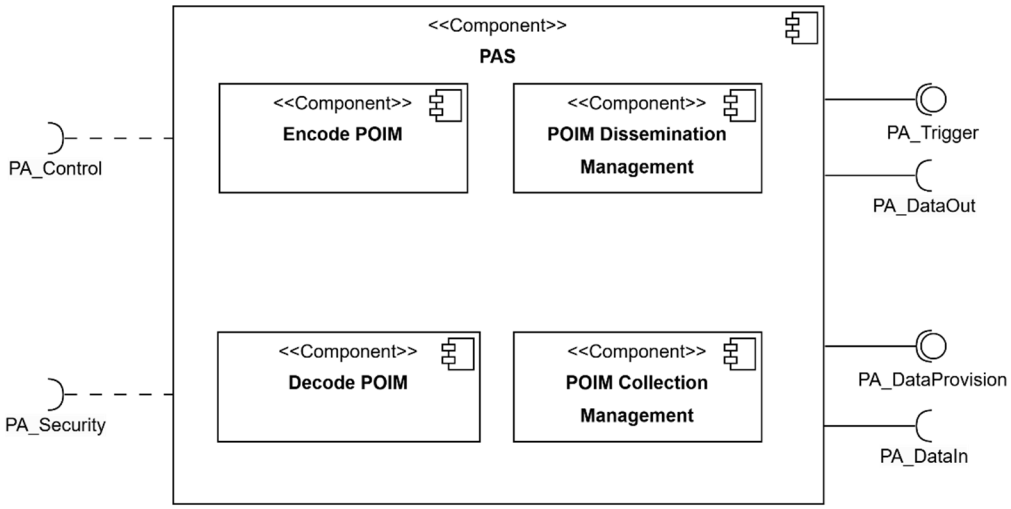


Figure 5: PAS component diagram

5.3 Interfaces of the PAS

5.3.1 Interface to the ITS-S applications

The PAS provides interfaces to applications for the processing of the payload at the transmitting ITS-S and the receiving ITS-S. An application may execute requests to the infrastructure services in the facilities layer to trigger, update or terminate dissemination of a message. In addition, a set of facilities control information is passed as specified in Table 1.

Table 1 presents the data contained in an application request.

Table 1: Interfaces to ITS-S applications

Category	Data	Definition
Data passed from application to disseminating PAS (PA_Trigger)	POIM set identification	Identification of the POIM set (only in case of update or ending).
	Request type	Trigger, update or ending of dissemination.
	Dissemination parameter	Conditional PCI and SCI data for PA_DataOut, see Table 2.
	Payload	Information to be contained in payload.
Data returned from disseminating PAS to the requesting application (PA_Trigger)	POIM set identification	Identification of the POIM set.
	Failure notification	Failure notification to the requesting application in case the request could not be processed correctly.
Data provided from collecting PAS to application (PA_DataProvision)	Payload	Parking availability information contained in POIM as payload.
	Collection parameter	Conditional PCI and SCI data from PA_DataIn, see Table 2.

5.3.2 Interface to management plan entities

The PAS may exchange information with the entity(ies) in the ITS management plane via the interface PA_Control.

NOTE: The specifications of the interface between the PAS and the management entity is out of scope of the present document.

5.3.3 Interface to security plan entities

In case ETSI ITS security (ETSI TS 103 097 [2]) at the facilities layer is used, the PAS exchanges information directly with the entity(ies) in the security plane (see Figure 2) via the interface PA_Security.

NOTE: The specifications of the interface between the PAS and the security entity is out of scope of the present document.

5.3.4 Interface to PA_DataIn and PA_DataOut

The PAS shall pass the POIM for dissemination to either another facilities layer entity such as Resource Management or a lower layer functionality such as the Networking & Transport Layer (NTL) as specified in Table 2. The PAS uses the PA_DataOut interface for this purpose.

The PAS shall gather POIMs from either another facility layer entity such as Resource Management (RM) or a lower layer functionality such as the NTL as specified in Table 1. The PAS uses PA_DataIn interface for this purpose.

Table 2: POIM exchanged over interfaces PA_DataIn and PA_DataOut

Category	Data	Data requirement	M/C (see note)	Remark/Condition
Data provided by the PAS via PA_DataOut	POIM	*{POIM}* as specified in Annex A.	M	
	SCI	Additional control information related to security, such as ITS-AID and SSP, which shall be listed in the Authorization Ticket associated to the private key used to sign the message at the NTL.	C	If a security scheme (e.g. ETSI security [2]) at NTL requiring SCI is used.
	PCI	Additional control information depending on the protocol stack applied in the NTL.	C	If a NTL protocol requiring PCI is used.
	RMI	Additional control information related to Resource Management.	C	if RM is available in the ITS-S and the default parameters have not been configured or need to be overwritten for a specific POIM.
Data gathered by the PAS via PA_DataIn	Received POIM	*{POIM}* as specified in Annex A.	M	
	SCI	Additional control information related to security, such as ITS-AID and SSP that are listed in the Authorization Ticket attached to the message, and the result of the security check at NTL.	C	If a security scheme (e.g. ETSI security [2]) at NTL requiring SCI is used.
	PCI	Additional control information depending on the protocol stack applied in the NTL.	C	If a NTL protocol requiring PCI is used.
NOTE: M/C are as follows: "M" indicates that the data shall always be included. "C" indicates that the data shall be included only if the condition is satisfied.				

The PAS may request and receive data from the PoTi service using the PA_DataIn.

NOTE 1: The specifications of the interface between the PAS and the PoTi service is out of scope of the present document.

The PAS may provide POIMs to the LDM using the PA_DataOut.

NOTE 2: The specifications of the interface between the PAS and the LDM is out of scope of the present document.

6 POIM-PA dissemination

6.1 POIM-PA dissemination concepts

6.1.1 POIM-PA dissemination requirements

POIM-PA dissemination can make use of point-to-point or point-to-multipoint communication, using single-hop or multi-hop mechanisms, depending on the ecosystem it is used in. The specification of the dissemination requirements is out of scope of the present document and can be found in the corresponding system profiles.

All POIM-PA of a set defined in clause 6.1.3 shall be transmitted over the same channel.

6.1.2 Inclusion of POI information blocks

The information provided to PAS by the ITS-S Application in its request (see Table 1) shall be included into POI Information Blocks of type *ParkingAvailabilityBlock* a specified in clause 6.1.3, clause 7 and Annex A. This information should apply to one single parking place.

POI information blocks of other types may be included into the POIM-PA if the content is provided by the ITS-S application. This information should be semantically related to the content of the POIM-PA set.

EXAMPLE 1: A POI information block containing departure times of public transport at a park and ride facility.

EXAMPLE 2: A POI information block containing information about electric vehicle charging station at a parking facility.

6.1.3 POIM-PA assembly

The size of a single ASN.1 UPER encoded POIM-PA shall not exceed the Maximum Transmission Unit (MTU) *MTU_POIM*, which corresponds to the access layer technology *MTU_AL* of the channels over which the POIMs are transported, reduced by the header size of the networking and transport layer protocol *HD_NT*:

$$SIZE(POIM) \leq MTU_POIM = MTU_AL - HD_NT$$

The POIM-PA assembly process is responsible for assembling the message components into a POIM-PA set consisting of zero, one or more POIM-PAs, ready for dissemination (FL-Assembly). The PAS shall identify the entire set using an internal identifier to be used over the interface PA_Trigger (see Table 1) for update and termination.

Each POIM-PA in the set shall contain the *ParkingAvailabilityBlock* as specified in clause 7 and with the following restrictions:

- As many instances of *ParkingArea* inside the component *capabilities* of type *ParkingSupplyCapabilitiesContainer* shall be included as possible whilst the UPER size of the POIM fulfils the size requirement above.
- The insertion of specific *ParkingArea* into a certain *ParkingAvailabilityBlock* identified by a *blockIdentificationNumber* shall remain the same throughout updates of the POIM-PA set.
- The POIM-PA shall contain all identifiers of related POI blocks of the POIM-PA set instantiated for Fl-Assembly.
- If possible, the POIM-PA may also contain other POI information block (see clause 6.1.2).

6.1.4 Trigger, update, repetition and termination of dissemination

The application triggers the PAS for the dissemination of the set of POIM-PA. The application provides all information to be included in the POIM-PA payload. The PAS assembled the information and constructs all POIM-PA of the set and passes them for dissemination as described in clause 5.

NOTE: The order of dissemination of the POIM-PA is irrelevant since the content of the various messages does not have a different priority or value of information.

The POIM-PA dissemination shall be repeated as requested until an update is provided by the application: in that case the current dissemination shall be terminated, and the new set of POIM-PA shall be constructed and disseminated.

The POIM-PA dissemination shall be terminated if the ITS-S application requests the termination.

6.2 POIM-PA dissemination constraints

6.2.1 Security constraints

6.2.1.1 Introduction

Clause 6.2.1 is applicable to ITS stations that are part of an ecosystem that uses the trust model according to ETSI TS 102 940 [4] and ITS certificates according to ETSI TS 103 097 [2]. The security mechanisms for ITS consider the authentication of messages transferred between ITS-Ss with certificates. A certificate indicates its holder's permissions, i.e. what statements the holder is allowed to make or privileges it is allowed to assert in a message signed by that certificate. The format for the certificates is specified in ETSI TS 103 097 [2]. Permissions are indicated by a pair of identifiers within the certificate, the ITS-AID and the SSP.

The ITS-Application Identifier (ITS-AID) indicates the overall type of permissions being granted: there is an ITS-AID that indicates that the originating ITS-S is entitled to operate a POI service to send POIMs, see ETSI TS 102 965 [6].

The Service Specific Permissions (SSP) is a field that indicates specific sets of permissions within the overall permissions indicated by the ITS-AID: for example, there may be an SSP value associated with the ITS-AID for POIM that indicates the originating ITS-S is entitled to send POIMs with a cause code value (defined in clause 7.1.4) set.

ITS-S provides SSP information in its certificate for all generated, signed POIMs. A received signed POIM is accepted by the receiving ITS-S if the POIM is consistent with the ITS-AID and SSP in its certificate.

6.2.1.2 Service Specific Permission (SSP)

POIMs shall be signed using private keys associated to Authorization Tickets of type explicit that shall contain a pair of: ITS-AID associated to the POI service as specified in ETSI TS 102 965 [6] and SSPs of type *BitmapSsp*, as specified in ETSI TS 103 097 [2]. The *BitmapSsp* for POIMs shall conform to the POIM SSP octet scheme illustrated in Figure 6. This octet scheme allows SSP format to accommodate current and future versions of the present document.

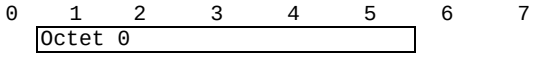


Figure 6: Format for SSP Octet Scheme (BitmapSsp)

The SSP has a maximum length as specified in ETSI TS 103 097 [2]. The first octet shall reflect the version of the SSP (see Table 3). In the current version of the present document, the SSP field contains only the SSP version byte.

Table 3: Octet Scheme for POIM SSPs

Octet #	Description	Value
0	SSP version control	1

6.2.2 General priority constraints

Priority information is provided in the PCI across the OSI layers and/or transmitted by lower layers.

7 POIM-PA Specification

7.1 POIM-PA structure

7.1.1 General Structure of a POIM-PA PDU

The message used by PAS belongs to the POIM class of messages and as such shall comply with the POIM PDU generic structure which is specified in ETSI TS 104 258 [5] and in Annex A. Therefore, the POIM is instantiated as a POIM for Parking Availability, abbreviated as POIM-PA. Information related to a parking place may be assembled into more than one POIM-PA if this is necessary due to MTU restrictions in transmission.

A POIM-PA shall contain one *ParkingAvailabilityBlock* and may contain other blocks that will be defined in later versions of the present document or in related documents such as standards for the availability of electric charging, generic energy supply or public transport facilities. The *ParkingAvailabilityBlock* is composed of containers (mandatory and optional) which transport information (data elements and data frames) being considered necessary for the receiving ITS-S to select an appropriate parking and act (e.g. reservation) to secure a parking space.

The general structure of the POIM-PA PDU, i.e. the *ParkingAvailabilityBlock* is illustrated in Figure 7. Each container is composed of a sequence of components, their component type being either a Data Element (DE) or a Data Frame (DF). A component is either optional or mandatory in the POIM Format. If not specified as optional in Annex A of the present document, a component is considered as mandatory. Components that are optional shall be present if the information is provided by the ITS-S application of the originating ITS-S.

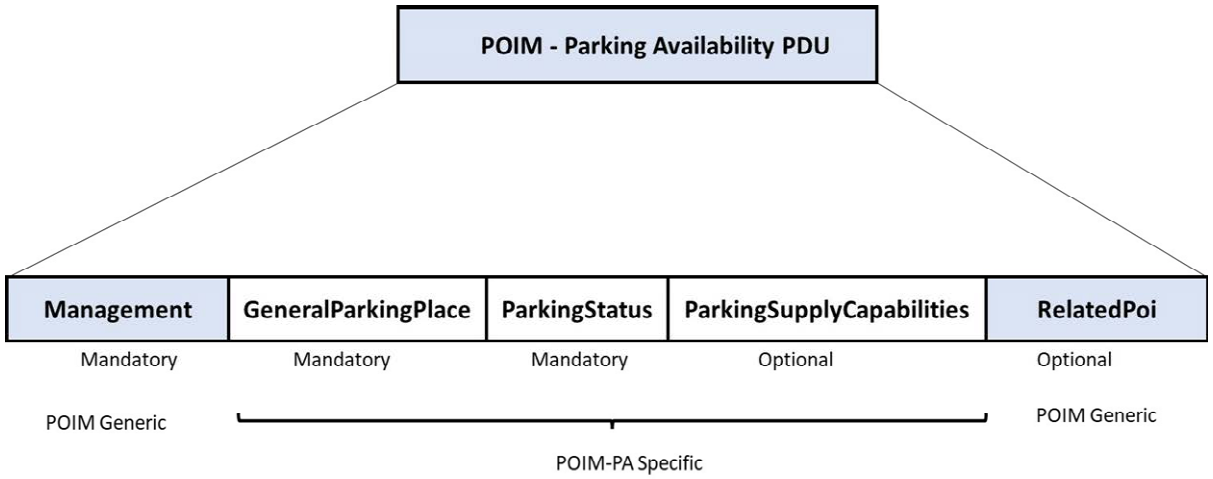


Figure 7: General structure of the POIM-PA PDU

The **Management** container is a generic POIM container is a mandatory container and contains data elements which are specified in clause 7.1.3.

The **General Parking Place** container is a mandatory container and contains general information about the parking place which are specified in clause 7.1.4.

The **Parking Status** container is mandatory and contains an aggregated status of the parking place which is specified in clause 7.1.5.

The **Parking Supply Capabilities** container is optional and contains a detailed status of available parking supply capabilities which are specified in clause 7.1.6.

The **Related POI** container is optional and contains one or several related POI(s) links enabling the access to these related POI(s) if judged necessary by the service user. Related POI data elements are specified in clause 7.1.7.

NOTE: The specification of protocol and message for accessing a related POI are out of the scope of the present document.

7.1.2 ITS PDU header

The ITS PDU header shall be as specified in ETSI TS 102 894-2 [1] and as constrained in Annex A.

7.1.3 Management container

The *ParkingAvailabilityBlock* component *managementContainer* represents the Management container and shall be of type *PoiBlockManagementContainer* as specified in ETSI TS 104 258 [5].

All POIM-PAs assembled as part of a set shall indicate the same value of the component *timestamp*.

The POIM-PA identification is enabled by the component *ParkingAvailabilityBlock.managementContainer.blockIdentificationNumber*. Each time a new POIM-PA set is generated upon an application request, a new set of *blockIdentificationNumber* values shall be assigned. A POIM-PA set shall use consecutive *blockIdentificationNumber* values for all POIM-PA in that set.

The *blockIdentificationNumber* shall uniquely identify the content of the *ParkingAvailabilityBlock* throughout various updates (see clause 6.1.3).

The identification of other messages that contain content linked to the content of the POIM-PA is enabled by the components *ParkingAvailabilityBlock.managementContainer.IvimReferences* and *ParkingAvailabilityBlock.managementContainer.MapReferences:*

- The component *linkedIvims* may be present if specific signage information applicable to the access roads to the parking facility or to the facility itself is conveyed using IVIM.

EXAMPLE 1: An IVIM access restriction to heavy goods vehicles only, applicable to the parking facility.

- The component *linkedMapems* may be present if the topology of the parking place and/or its access road are conveyed using MAPEM.

EXAMPLE 2: An MAPEM provides information about the access road and the main roads inside the parking facility.

7.1.4 General Parking Place container

The *ParkingAvailabilityBlock* component *placeInfo* represents the General Parking Place container and shall be of type *GeneralParkingPlaceContainer* as specified in Annex A. This container provides information regarding the parking place.

7.1.5 Parking Status container

The *ParkingAvailabilityBlock* component *aggregatedStatus* represents the Parking Status container and shall be of type *ParkingStatusContainer* as specified in Annex A. This container provides aggregated dynamic information relatively to the status of the parking place.

Each POIM-PA in a set shall contain the *ParkingAvailabilityBlock.aggregatedStatus* with the following recommendations:

- If the POIM-PA set consists of 1 POIM-PA the component *detailedStatus-Block* should be absent and the component *detailedStatus-Place* of *ParkingStatusContainer* may be present and if present it shall be used according to the requirements in Annex A.

- If the POIM-PA set consists of more than 1 POIM-PA the components *detailedStatus-Block* and *detailedStatus-Place* of *ParkingStatusContainer* should both be present and if present shall be used according to the requirements in Annex A.

The components *detailedStatus-Block* and *detailedStatus-Place* are both of type *ParkingStatusList* as specified in Annex A. Its subtype *ParkingStatus* can be tailored to specific needs to convey the aggregated (i.e. "sum") of the occupancy for a certain category/type of parking by specifying the logical AND combination of parking components as specified in Annex A.

EXAMPLE 1: The aggregated occupancy for parking vehicles of type "car" for "pregnant" persons.

EXAMPLE 2: The aggregated occupancy for parking vehicles of type "truck" with "refrigerated goods".

The *occupancy* information is provided in terms of occupied and free spaces, together with a rate in percentage and an evolution trend.

7.1.6 Parking Supply Capabilities container

The *ParkingAvailabilityBlock* component *capabilities* represents the Parking Supply Capabilities Container and shall be of type *ParkingSupplyCapabilitiesContainer* as specified in Annex A. If present, the component shall contain information related to one or more parking areas of the parking place according to one of the following mutually exclusive options:

- Default: All parking areas of a parking place shall be listed each as single DF *ParkingArea*.
- Application selection: Only specific parking areas of a parking place shall be listed each as single DF *ParkingArea*, e.g. only the truck parking areas.
- Facilities layer assembly (see clause 6.1.3): only selected parking areas of a parking place shall be listed each as single DF *ParkingArea*, i.e. only those that fit into one POIM-PA according to the MTU size.

7.1.7 Related POI container

The *ParkingAvailabilityBlock* component *relatedPoi* represents the Related POI container and shall be of type *RelatedPoiInfoBlockContainer* as specified in ETSI TS 104 258 [5]. If present, the component shall contain information related to other POI information blocks according to one of the following options (combinations are allowed):

- Facilities layer assembly (see clause 6.1.3): references to other POI information blocks of the same type (parking availability), that carry information related to parking areas of the same parking place, shall be listed each as a single DF *RelatedPoiInfoBlock*.
- Cross-block referencing: references to other POI information blocks of other types physically related to the parking place, shall be listed each as a single DF *RelatedPoiInfoBlock*.

The identification of related POI information blocks generated by PAS for Facilities layer assembly is enabled by the component *ParkingAvailabilityBlock.relatedPoi* with values of *ParkingAvailabilityBlock.relatedPoi.poiInfoBlockType = 1*.

The identification of related POI information blocks generated by other facilities layer services is enabled by the component *ParkingAvailabilityBlock.relatedPoi* with values of *ParkingAvailabilityBlock.relatedPoi.poiInfoBlockType* $\neq$ *1*. These POI Information blocks may be present inside a POIM-PA of the same set (see clause 6.1.2) or in other POIM from other facilities layer services.

EXAMPLE 1: A POI information block containing departure times of public transport at a park and ride facility.

EXAMPLE 2: A POI information block containing information about electric vehicle charging station at a parking facility.

7.2 POIM-PA format

The POIM-PA syntax and semantics shall be as specified in Annex A of the present document.

DEs and DFs that are not defined in the present document shall be imported from the ETSI Common Data Dictionary ETSI TS 102 894-2 [1] as specified in annex A.

Unaligned Packet Encoding Rules as defined in Recommendation ITU-T X.691/ISO/IEC 8825-2 [3] shall be used from POIM-PA encoding and decoding.

Annex A (normative): ASN.1 modules

This clause provides the normative ASN.1 module containing the syntactical specification of the *ParkingAvailabilityBlock*, its containers, data frames, and data elements defined in the present document.

The semantical specification of the *ParkingAvailabilityBlock* components, its containers, the data frames, and the data elements are contained in the same module, in the form of ASN.1 comments and is also available in readable format as indicated in Table A.1.

The POIM-ParkingAvailability module is identified by the Object Identifier {itu-t (0) identified-organization (4) etsi (0) itsDomain (5) wg1 (1) poi (103916) parkingAvailability (1) major-version-2 (2) minor-version-1 (1)}. The module can be downloaded as a file as indicated in Table A.1. The associated SHA-256 cryptographic hash digest of the referenced file offers a mean to verify the integrity of that file.

Table A.1: POIM ParkingAvailability ASN.1 module information

Module Name	POIM-ParkingAvailability
OID	{itu-t (0) identified-organization (4) etsi (0) itsDomain (5) wg1 (1) poi (103916) parkingAvailability (1) major-version-2 (2) minor-version-1 (1)}
Link	https://forge.etsi.org/rep/ITS/asn1/pa_ts103916/-/raw/v2.2.1/POIM-ParkingAvailability.asn
SHA-256 hash	f54c6860776a7ce25721420bb7a9fec51d5f955da3c07d559f5034905a2e3687
Readable format	https://forge.etsi.org/rep/ITS/asn1/pa_ts103916/-/blob/v2.2.1/docs/POIM-ParkingAvailability.md

This module is intended to be compiled together with the modules specified in ETSI TS 104 258 [5] Annex A. The following additions shall be made to the module POIM-ExternalModule (see example in the ASN.1 comments):

- The value assignment of the value *parkingAvailability* to *PoiType* value 1.
- The import of the data type *ParkingAvailabilityBlock* from the module POIM-ParkingAvailability.
- The addition of the information object *ParkingAvailabilityBlock* identified by the value *parkingAvailability*.

History

Version	Date	Status
V2.1.1	January 2024	Publication
V2.2.1	October 2026	Publication